

Soliton Interview Experience

Salary Package - 10 LPA

Round 1 - Preliminary MCQ's & Programming (Online)

- Round 1 went for an hour.
- The first preliminary round is for MCQ's that is for 1 hour which consists of questions from Physics and Programming.
(programming-basic c code(finding output), physics - electricity, boolean logic dates, kinetics, mechanics, force , gravitation etc)
- They want you to use the rough sheet to work on your problems. These rough works play a major role in the interview process.
- Out of 400 candidates 150 got shortlisted
 1. Questions based on strings and pointers in C.
 2. Pattern logic (a sequence of numbers will be given and you have to predict the next number).
 3. Blood relation.
 4. Physics problems in the topic of Kinematics and Circuit theories.

Sample Question 1:

```
#include <stdio.h>
int main(){
    char string[] = "Examly";
    printf("%s", string+2);
    return 0;
}
```

Answer: amly

Sample Questions 2:

```
#include <stdio.h>
int main(){
    int arr[] = {1, 2, 3, 4, 5, 6};
    int *ptr = (&arr)+1;
    printf("%d", *(ptr-1));
    return 0;
}
```

Answer: 6

Sample Question 3:

```
#include <stdio.h>
int main(){
    int matrix[3][2] = {6, 8, 13, 11, 5, 4};
```

```

for(int i = 0; i < 3; i++){
    for(int j = 0; j < 2; j++){
        printf("%d ", matrix[i][j]);
    }
}
return 0;
}

```

Answer : 6 8 13 11 5 4

Sample Question 4:

What is the default value of an integer array when left uninitialized?

Answer: 0

Section 2 – Maths & Logical

Questions in Maths were based on probability, integrals, differentiation, AP.

Sample Question 1:

A police officer fires three bullets on a thief. The probability of the thief getting killed by one bullet is 0.8. Find the probability that the thief is alive.

Answer: 0.008

$P(\text{Killed by one bullet}) = 0.8$

$P(\text{Alive by one bullet}) = 1 - 0.8 = 0.2$

For three bullets, $P(\text{Alive for three bullets}) = 0.2 * 0.2 * 0.2 = 0.008$

Sample Question 2:

221, 220, 223, 222, 225, 224,....

What are the next two numbers will come next in the given series?

Answer: 227, 226

Explanation: Series of shuffled odd and even numbers

Sample Question 3:

Given a function $f(x) = x - 1/x$, find $f'(1)$.

Answer : 2

Explanation:

$f(x) = (x^2 - 1)/x$

$f'(x) = (2x - 0)/x$

If x is -1, $f'(x) = -2/-1 = 2$

Another question was to find the result of integration.

Questions on logical thinking were based on Blood Relations, Decoding words.

Sample Question 1:

A man said to a woman seeing a picture, 'His mother is the only daughter of your father'. How is the woman related to the person in the picture?

Answer: Mother

Sample Question 2:

If TWELVE is represented in a language as 852132 and NINE is represented as 9792. How will you represent ELEVEN in that language?

Answer: 212329

Explanation: T – 8, W – 5, E – 2, L – 1, V – 3, N – 9, I – 7

ELEVEN – 212329

Section 3 - Physics & Technical

Questions were based on current electricity, conservation of momentum, speed, kinetic and potential energy, logic gates. Three questions were based on the only formula $V = IR$.

Sample Question 1:

A current of 33 A flows through a circuit for 0.5 seconds. What is the value of charge flowing through the circuit?

Answer : 66 C

Explanation: $I = Q/t = 33 / 0.5 = 66 \text{ C}$

Sample Question 2:

An iron box is connected to voltage source of 220 V and consumes 0.5 A current. If the voltage is reduced to 120 V. Find the current.

Answer : 0.27 A

Explanation: $R = V / I = 220 / 0.5 = 440 \text{ ohm}$

$V = 120 \text{ V}, I = V / R = 120 / 440 = 0.27 \text{ A}$

Sample Question 3:

An object is moving at a velocity of 20 m/s. If it is thrown upwards, what is the maximum height it can reach? Take $g = 9.8 \text{ m/s}^2$

Answer: 20 m

Explanation: Kinetic Energy = Potential Energy

$\frac{1}{2}(mv^2) = mgh$ (Cancel m in both sides)

$\frac{1}{2}(20^2) = 10 * h$

$200 = 10h$

$h = 20 \text{ m/s}$

Sample Question 4:

A car of mass 1500 kg is moving with a velocity of 80 km/h strikes a car of mass 1000 kg moving at a velocity of 30 km/h travelling in the same direction. After collision, what will be the resultant velocities of the cars?

Answer: 60 km/h

Explanation: By conservation of momentum, $m_1v_1 + m_2v_2 = (m_1+m_2) v$

$1500 * 80 + 1000 * 30 = (2500) v$

$v = (120000 + 30000)/2500 = 60 \text{ km/h}$

Sample Question 4: What happens when a capacitor gets fully charged?

Two questions were based on logic circuits, one was to find what gate it is, another question was to find the values of a, b, c to get the given output.

Round 2 - Soliton Test - Fill in the blanks (Online)

- This round goes for an hour. It consists of 10 questions of 1 mark each.
- Questions are divided into Aptitude (2 questions), Physics & Core (5 questions) and Programming Output (3 questions).
- Formulae will be given in the beginning. (only basic formulas)
- No MCQ questions only fill up.
 1. Signal conversion .
 2. Amount of paint required to fill a 3D figure.

3. Basic integration and differentiation questions.
4. Questions based on circuit theory

Sample Question 1.1

A function is said to be an odd function when $f(x) = -f(-x)$. For the function $f(x) = x^2 - nx + 4$ to be even, find the value of n .

Answer: 0

Explanation: For even function, $f(x) = f(-x)$

$$f(1) = 1 - n - 4 = 5 - n$$

$$f(-1) = 1 + n + 5 = 5 + n$$

$$5 - n = 5 + n$$

$$5 - 5 = n + n = 2n$$

$$n = 0/2 = 0$$

Sample Question 1.2 [Figure will be given – it's like a plus sign]

A shape as shown in the picture is formed by joining five cubes of side 2 cm. The cube is painted on its outer side as shown. If each square cm requires 2 ml of paint. Find the total amount of paint required to paint the shape formed.

Answer: 96 ml

Explanation: Three sides of each cube were visibly painted, front and back were not painted, One side of each cube is facing the centre cube, so not painted. So, four cubes are painted, centre cube is not painted.

$$\text{Area of 1 side of the cube} = a^2 = 2 * 2 \text{ sq cm} = 4 \text{ sq cm}$$

$$\text{Area of 12 sides} = 12 * 4 = 48 \text{ sq cm}$$

$$\text{Paint required for 1 sq cm} = 2 \text{ ml}$$

$$\text{Paint required for 48 sq cm} = 2 * 48 = 96 \text{ ml}$$

Sample Question 2.1

Given the equation for work done (W) and displacement (s) in terms of t , find the power.

Explanation: $P = dW/dt$

Find the value of t from displacement equation. Differentiate the equation of W , substitute the value of t obtained from equation of s .

Sample Question 2.2

Given a circuit with four resistors connected to a source, find the power. $R_1 = R_2 = R_3 = 3 \text{ ohms}$, $R_4 = 0.5 \text{ ohm}$

Sample Question 2.3

Given two input waveforms and an output waveform, find the operation to be performed on the input waves to get the resultant waveform.

Sample Question 2.4

Given a logic gate circuit, find the total time duration of HIGH(1) output signal.

Answer: 5

Explanation: The equation of the output can be found from the circuit, I got it as, $Y = (A+B).(B+C)$, so I got 1 for five of the eight combinations, so answered it as 5.

Sample Question 3.1

Find the operation that is being performed in the given flowchart. Flow chart will be given.

Answer: Finding factorial of input number N and printing the result

Sample Question 3.2

Find the output of the given program.

```
#include <stdio.h>
int main() {
    int n = 0, i;
    for(i = 0; i < 3; i++){
        switch(i){
            case 0:
                n -= i;
            case 1:
                n += i;
            case 2:
                n *= i;
                break;
            default:
                n += 5;
                break;
        }
    }
    printf("%d %d", n, i);
    return 0;
}
```

Answer: 2 3

Explanation: Remember that break is applicable only inside loops and switch. I answered this question wrong.

Sample Question 3.3

Find the output of the given program. The question was with string and pointers. The string is "COMPILE", one pointer sp1 is at the start of the character array, another pointer sp2 was at the end. Loop through the length of the string, and inside the loop assign *sp1 to *sp2, also string[i] = *sp1.

Answer: COMPMOC

Round 3 - Programming Round I (Online)

- This round goes for an hour. It consists of 3 questions. It's not necessary to solve all problems but trying to solve it will be better before skipping to the next question.
- They expect us to complete at least 1 Question. There's no way to switch to previous question.
- Out of 40 candidates 25 got shortlisted.
- Below are the questions asked:
 1. Reverse a string.
 2. Given an array of numbers, merge the numbers into a single number. i.e (1 , 2 , 3 -> 123).
 3. Valid parentheses (Leetcode #20)

Sample Questions :

1. Given an encrypted string, print the resultant decrypted string (String reversal)

Input: notiloS ta krow I

Output: I work at Soliton

Approach: It's simply string reversal, but try not to use `strrev`, you may use `strlen`. The code logic is also important, not just the output.

2. Given the elements of an array `A[]` convert it to an integer, add 1 to the result and print. The most significant digit is moved to the first element of the array.

Input: 1 2 3

Output: 124

Input: 0 8

Output: 9

Approach: Array size will not be given so initialize array of size of your choice and then get inputs and perform operation, `result += result * 10 + array[i]`.

3. Given a string containing brackets `(){}[]` check whether it is balanced or not. If some other characters are present, it is considered invalid.

Input: `[{}]()`

Output: Balanced

Input: `([]{}-`

Output: Invalid

Approach: You may solve it with if else. Initialize variables to count parentheses, `flower_bracket`, `square_bracket`, `invalid` to 0. Traverse the string, when open bracket is there, increment the corresponding variable, when close bracket is there, decrement the variable. If character is invalid, set `invalid` to 1. Come out of the loop, check whether all the variables are still having value 0 using `&&` and print the result accordingly.

Round 4 - Personal HR (Offline):

- Self-introduction (Schooling and Where do you come from?)
- Why Soliton?
- Any questions do you want to ask me?

Round 5 - Technical HR (Offline):

- They ask you to explain the approach you did to solve a particular problem in
- Questions on our area of Interest.
- Questions from previous rounds attended.
- Any questions do you want to ask me?
- Round 4 and 5 held parallel. Out of 25 candidates 15 got shortlisted.

Round 6 - Programming Round II (Long Programming Round) (Online)

- This Programming round goes for 3hr 30 mins.

- There were three questions and before attempting any question the logic should be explained to the technical team of Soliton.
- If they are satisfied with our logic, we can start coding our logic.
- Username and Password validation like in a database.
- Maximum area of a triangle in a grid. <https://www.geeksforgeeks.org/maximum-area-triangle-different-vertex-colors/>
- The most important part is communicating our logic to the technical team.
- Out of 15 candidates 10 got shortlisted.

Sample Questions :

Sample Question 1:

Given the username and password, check whether it will be accepted or rejected. The password length should be at least 8 should contain at least one capital letter, one small letter, one special character, one number. The allowed special characters are @#\$%^&*.

The username may contain alphanumeric characters and optionally may also have @ or #.

If the username is already taken, it is rejected.

Input:

soliton@1 : Password@1

soliton#2 : password—

soliton#2 : Abcd123@

Output:

Accepted : Accepted

Accepted : Rejected

Rejected : Accepted

Sample Question 2:

Given array elements, find the starting index and ending index of the minimum sub array which on sorting makes the complete array as sorted. Consider index starts with 1. If the array is already sorted, return -1 and print the original array.

First line of the output contains the starting index, and ending index of the minimum sub array to be sorted.

Second line of the output is the sorted array.

Input: 1 3 2 4 5

Output: [2, 3]

[1 2 3 4 5]

Input: 1 2 3 4 5

Output: -1

[1 2 3 4 5]

Sample Question 3:

GeekForGeeks – Maximum area of triangle having different vertex colors

Round 7 - Design Round (Offline):

- This round is similar to Group Discussion, but with some differences.
- They divided us into a batch of 5 and gave a problem statement for which we should discuss among ourselves to get an optimal solution for the problem.
- Here is the question,
Design an optimal solution to maintain a constant voltage across all cells in a battery.
- We should discuss and express our design first, then we should present our design to others using a whiteboard.
- **Battery Management System** – Design a system to maintain equal voltage across all the cells of a battery
- We were divided into two groups, five in each. First, individually we need to think of the solution and write it down in a paper. We can get help from the technical people from Soliton who will be with us. After that, we need to explain our solution in the white board individually. The next is to discuss about the solution as a group. After discussion, again write the optimized design/algorithm/flowchart/pseudo code in paper.
- Out of 10 candidates 5 got shortlisted.

Round 8 - Essay Writing (Offline):

- Describe Yourself.
- Why Soliton?
- What Soliton will do to you and what you will do to soliton?
- What is your ambition?
- Where do you see yourself in 5 years, both Professional and Personal.
- This round is not much of a deciding round , feel free to express your personal thoughts.
- Out of 5 candidates all 5 of us got the offer.

Points to be noted:

1. Programming to be done only in C.
2. Go through the basics of 12/11th Physics.
3. Concentrate on Arrays, Strings, Pointers .
4. Be confident and communicate boldly.
5. Mention only the topics you know in the resume.

